

DBMS Lab 1

Kunal Dhanesh Soni D6EC 57

SELECT * FROM Customers;

23458	Gupta	Vikas	51, Station Road	Jaipur	NULL	NULL
34568	Desai	Priya	19, Andheri East	Mumbai	NULL	NULL
42869	YAHOO	NITIN	TEL AVIV	JERUSILAM	NULL	NULL
45678	Rumar	Anil	34, Brigade Road	Bangalore	NULL	NULL
58928	Mehta	Neha	8, FC Road	Pune	NULL	NULL
67238	Patel	Rahul	15, Park Street	Kolkata	NULL	NULL
69428	TRUMP	MCDONALD	WASHINGTON	USAAA	NULL	NULL
78458	Sharwa	Amit	22, MG Road	Delhi	NULL	NULL
89798	Joshi	Ramesh	413, Sadashiv Nagar	Pune	ramesh.joshi@abc.com	Regular Buyer
90878	Nair	Arjun	12, Marine Drive	Mumbai	NULL	NULL
96308	Singh	Prakesh	87, House 10, abc lane	Mumbai	NULL	NULL

SELECT CustomerID, Address FROM Customers WHERE CustomerID=96300;

DESC Customers;

SELECT * FROM Customers WHERE CustomerID=96300;

What is a database? Why do we need a database management system such as MySQL?

A database is an organized collection of data that can be stored, accessed, and managed easily. For example, a customer database may store customer names, IDs, phone numbers, and addresses.

A Database Management System (DBMS) such as MySQL is software used to create, store, retrieve, update, and manage data in databases. We need a DBMS because it makes handling large amounts of data easier, faster, more secure, and more organized.

What is the difference between CREATE DATABASE, SHOW DATABASES, and DROP DATABASE?

Command	Purpose
CREATE DATABASE	Creates a new database.
SHOW DATABASES	Displays the databases available in MySQL.
DROP DATABASE	Deletes an entire database and its contents.

What is a table in a database? Explain the purpose of rows and columns in a table.

A table is a structured collection of related data in a database. It consists of rows and columns.

- Columns represent the different attributes or fields of the data.
- Rows represent individual records.

What is a primary key? Why was CustomerID selected as the primary key in the Customers table?

A **primary key** is a column, or combination of columns, that **uniquely identifies each row** in a table. A primary key cannot contain duplicate values and normally cannot be **NULL**.

CustomerID is a suitable primary key for the **Customers** table because each customer can be assigned a **unique ID**. Unlike names or addresses, customer IDs can reliably distinguish one customer from another.

What is the purpose of the VARCHAR and INT data types? Give one suitable example of each.

VARCHAR is used to store variable-length text/string values.

Example:

Name VARCHAR(50)

This can store a customer's name, such as "**Rahul Sharma**".

INT is used to store whole numbers (integers).

Example:

Age INT

This can store a value such as **25**.

What is the difference between INSERT, SELECT, UPDATE, and ALTER TABLE?

Command	Purpose
<u>INSERT</u>	Adds a new record/rows to a table.
<u>SELECT</u>	Retrieves or displays data from the table.
<u>UPDATE</u>	Changes the existing data in a table.
<u>ALTER TABLE</u>	Changes the structure of an existing table, such as adding or removing columns.

Why is the WHERE clause important in an UPDATE statement? What could happen if we write an UPDATE statement without a WHERE condition?

The **WHERE** clause specifies which rows should be updated . It is therefore extremely important when modifying existing data.

For example:

```
UPDATE Customers
```

```
SET Age = 26
```

```
WHERE CustomerID = 101;
```

Only the customer whose **CustomerID** is **101** will be updated.

If we write:

```
UPDATE Customers SET Age = 26;
```

without a WHERE condition, MySQL will update every row in the Customers table. This could accidentally change all customers' ages to **26**.

Therefore, when using **UPDATE**, always check that the **WHERE** condition identifies exactly the records you intend to change.